



Fluid flow simulation on a Turritella-seashell-like geometry demonstrating its ability as static mixer for inline mixing

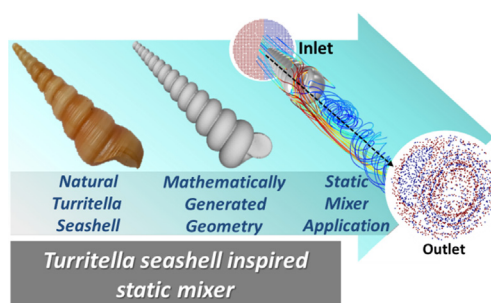
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HIGHLIGHTS

- Mathematically generated simplified turritella seashell geometry.
- Turritella geometry in a pipeline acts as a static mixer in turbulent flow regime.
- Two-step mechanism involving flow splitting and no-slip at pipe wall.
- Mixing entropy constant over a wide range of fluid flow velocity.
- Tunable insert parameters to reduce pressure drop without affecting mixing entropy.

GRAPHICAL ABSTRACT



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ABSTRACT

Inspired by the turritella's unique hydrodynamic behaviour of self-orientation parallel to the current and specialization in burrowing, here, we demonstrate that a turritella-seashell-like geometry can serve as an effective static mixer insert to induce inline mixing in a pipe. A method to mathematically parametrize a simplified turritella-seashell-like geometry is presented. Through CFD studies, a two-step mixing mechanism is proposed and its performance is investigated as a function of the inlet velocity, pipe diameter, surface boundary conditions and insert geometry parameters. The mixing output is nearly independent of the inlet velocity in the studied range. The no-slip boundary condition at the pipe wall is essential in creating shearing in the fluid, without which no mixing is observed. The insert geometry can be tuned to minimize the pressure drop by over 40% without significantly affecting the mixing output. The insights obtained herewith are discussed in interest of designing new static mixers.

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1. Introduction

Biomimicry has proven to be an invaluable paradigm shift in guiding innovations that are naturally functional and efficient across various fields. Designs have found inspiration anywhere from the aerodynamic beak of the kingfisher (Foo et al., 2017; Crandell et al., 2019) to the mosquito avoiding obstacles in poor light environments (Nakata et al., 2020). Adding to the platter of bio-inspiration, through this study, we demonstrate for the first

time that the geometry of the turritella seashell placed in a pipe induces fluid mixing and therefore exhibits potential as a static mixer application.

The turritella shell, also known as the 'common tower shell', is named after the speciose genus of turritella sea snails. The geometry of the shell, which roughly resembles the shape of a cornucopia, consists of a self-similar spiral that begins from the apex and grows outward in a highly twisted manner to form an elongated conical shape, see Fig. 1a. After the death of the sea snail, the hollow shells are swept ashore, where they are often inhabited by hermit crabs. A typical observation on the beach is that the turritella shell obstructs the incoming flow of the swash and creates undulating

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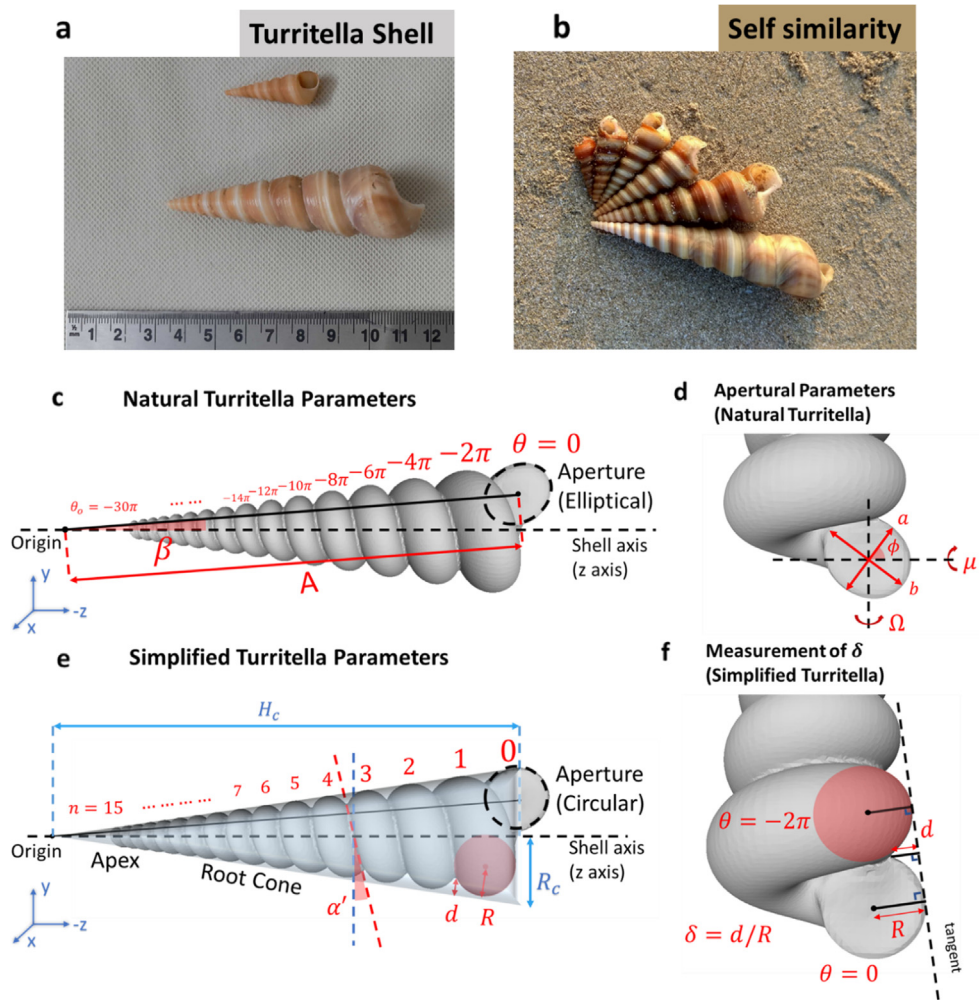


Fig. 1. (a) Comparison of two different sized turritella seashells alongside a scale. (b) Turritella seashells of different sizes exhibiting a self-similar spiral throughout the stages of their growth. (c) Depiction of parameters A , β and θ from the original 11-parameter model of the natural turritella. (d) Depiction of apertural parameters a , b , ϕ , μ and Ω from the original 11-parameter model of the natural turritella. (e) Description of the simplified turritella in terms of the 5 auxiliary parameters R_c , H_c , α , δ and n . (f) Depiction of the circular aperture in the simplified turritella and the measurement of the depth ratio of the whorls δ .

V-shaped patterns in the sand. Especially on beaches with laminated sand, darker grains beneath are exposed in sharp visual contrast to the lighter grains above, creating rich, undulating patterns that extend up to 1 or 2 m leeward of the shell. While this behaviour is typical to any object or shell in the sand, the turritella is particularly conspicuous among other objects or shells on the beach in that it orients its longest axis *parallel* to the current. Most other shells, on the other hand, are observed to orient their axis perpendicular to the current (Branchley and Newall, 1970). Further, the parallel orientation can be classified into a metastable apex down-current orientation and a stable apex up-current orientation. These two stable and metastable states have been observed in natural as well as experimental settings through many biological (Negus, 1968; Graham, 1938) and paleontological studies (Branchley and Newall, 1970; Stewart Nagle, 1967; Nebelsick et al., 2020; Scott, 1970). On the beach, as the swash comes in (deep enough to fully submerge the shell), the shell assumes an apex up-current orientation since it is the most streamlined configuration; and as the swash retreats and the flow reverses, the apex acts as a frictional anchor around which the shell skews 180° until the apex is up-current again (Branchley and Newall, 1970), see Supplementary Video 1. This parallel orientation is so reliably correlated with the flow direction that it is used to track the evolution of river flow across geological timescales by observing turritelline

fossils and their orientations (Nebelsick et al., 2020). Besides preferential orientation of the hollow turritella shell, species that find shelter in the turritella shell also demonstrate some intriguing behaviour. Inhabitant sea snails like the *Turritella communis* Risso are known to be highly specialized burrowers in gravelly mud (Graham, 1938; Yonge, 1946). Hermit crabs on the beach that take shelter in turritella seashells are observed to bury themselves with ease within one swash cycle, despite the long and bulky nature of the shell. Even hollow shells are observed to partially self-bury over multiple swash cycles on certain beaches. We suspect that shell-enhanced hydrodynamic scour formation plays a role in assisting the burrowing of both these species. Scour formation is observed when an object of any shape obstructs a unidirectional and oscillatory flow. This generates vortices or swirls that locally change the bottom stress, thereby enhancing the sediment transport rate (Inman and Jenkins, 2019; Jenkins et al., 2007). This bottom stress is a key factor in causing sediment transport. In the special case of the turritella shell, we hypothesize that the spiral whorls help in directing the flow into the region between the sandy surface and the shell, thereby enhancing flow constriction and sediment transport.

It is from here that we take an engineering viewpoint and ask the following question: If the effect of flow constriction between the sandy surface and the turritella shell can cause sediment trans-

port to assist burying, would it be possible to enhance this constriction effect if, instead, the turritella shell is bound on all sides by a surface? This is equivalent to placing the turritella inside a pipe, where instead of transporting grains from a surface, the high constriction may contribute to mixing particles in the fluid or mixing the fluid itself. An arrangement like this essentially serves the purpose of a static mixer. A static mixer is a device that modulates the behaviour of the fluid via its specialized geometry to mix the incoming fluid at the expense of pressure drop (Ghanem et al., 2014). Static mixers have been in the limelight of many process intensification developments in the chemical engineering industry as well as a burgeoning thrust area in academia in recent decades (Thakur et al., 2003). Among the benefits these devices have to offer, the salient ones include relatively low equipment operation and maintenance costs, sharp residence time distributions, modularity, and enhanced safety during operation. Considering the ability of the turritella shell geometry to cause sediment transport by modulating fluid flow behaviour, we explore a turritella-inspired arrangement on the lines of static mixer design.

Many comprehensive reviews have listed the available designs of static mixers and micromixers (Ghanem et al., 2014; Thakur et al., 2003; Hessel et al., 2005; Lee et al., 2016; Nguyen and Wu, 2005; Hessel et al., 2008; Lee et al., 2011; Yao et al., 2015; Hardt et al., 2005; Ghanem et al., 2013). Popular designs include: folding structures splitting and recombining flows, serpentine channels, and inserts increasing tortuosity. One method of classification of static mixers is based on their functional geometry (Khalde et al., 2020): a) Use of bends with a constant cross-section of channel/tube, b) Use of varying cross-section of channel/tube, c) Use of channel branching for flow splitting and recombination, and d) Use of inserts in the channel. The turritella-like static mixer we investigate in this CFD study is primarily of the fourth category but leverages the features of varying cross-sectional area and flow splitting of the second and third categories, respectively.

In this study, using computational fluid dynamics (CFD) simulations, we show that a turritella-like insert placed in a pipe can function as a static mixer. The range of operation of this turritella-in-a-pipe configuration is in the turbulent regime of Reynolds number (Re) 8000–60000, which corresponds to the regime observed in the natural setting of the turritella seashell on a typical beach. The flow regime on a typical beach is determined by calculating Re with a characteristic velocity equal to the incoming swash velocity (0.1–2 m/s) and the characteristic length equal to the depth of the swash layer (2–4 cm) (Guza et al., 1985). In this study, we first develop a method to construct and parametrize a simplified turritella-seashell-like geometry. We demonstrate the mixing capability and further prove that it is specific to the turritella seashell by comparing the turritella with other simple geometric inserts of similar sizes. A two-step mechanism is proposed by virtue of which the turritella insert creates mixing in the fluid: (a) flow splitting due to asymmetric constriction and (b) shearing due to the pipe wall. We derive evidence supporting the proposed mixing mechanism by studying the performance of this turritella-like static mixer as a function of the inlet velocity, pipe diameter and the slip condition at the geometry surfaces. We also investigate the effect of varying the turritella geometry on the mixing performance. Finally, we identify the key implications of using a turritella-like geometry as a static mixer and discuss how newer designs of static mixers (not necessarily turritella-shaped) might emulate the mixing mechanism of the turritella-like static mixer. We believe that the insights drawn from this study can assist in furthering research on turritella-like designs and in developing generic static mixer designs emulating the two-step mechanism.

2. Methods

2.1. Development of the simplified turritella geometry

The basic structure of a turritella shell constitutes a logarithmic, self-similar spiraling whorl winding around a central axis (Fig. 1a). Remarkably, the self-similar nature of the spiral is maintained throughout the inhabitant sea snail's lifetime as the shell grows (Cortie et al., 1992) (Fig. 1b). The number of spirals and aperture size increases as the mollusc grows and more layers are added. A remarkable feature of many molluscan shells is that the equiangular angle of the spiral remains constant throughout the life of the shell, resulting in the observed self-similar geometry (Cortie et al., 1992). This forms the basis for the development of the mathematical model of turritella seashell geometry.

For the present study, we develop a simplified 5-parameter model that captures the most salient features of the natural turritella. Here, we begin our derivation for the 5-parameter model, starting with the eleven parameters of Cortie's generalized model (Cortie et al., 1992; Cortie, 1989; Cortie, 1993) for molluscan shells. Cortie's model makes use of the logarithmic self-similar nature to describe 3D parametric surfaces of molluscan shells, given by:

$$x = D \cdot [A \sin \beta \cos \theta + R \cos (s + \phi) \cos (\theta + \Omega) - R \sin \mu \sin (s + \phi) \sin \theta] \cdot e^{\theta \cot \alpha} \quad (1)$$

$$y = [-A \sin \beta \sin \theta - R \cos (s + \phi) \sin (\theta + \Omega) - R \sin \mu \sin (s + \phi) \cos \theta] \cdot e^{\theta \cot \alpha} \quad (2)$$

$$z = [-A \cos \beta + R \sin (s + \phi) \cos \mu] \cdot e^{\theta \cot \alpha} \quad (3)$$

$$R = [a^2 (\cos s)^2 + b^2 (\sin s)^2]^{1/2} \quad (4)$$

Here, x , y , and z are 3D cartesian coordinates, each of which is a function of the parametric coordinates s and θ . As s and θ are varied, the initial elliptical aperture (Fig. 1c) is rotated and translated towards the origin in a spiral fashion about a central axis (shell axis/ z -axis) to generate the whorls of the shell. The size of the aperture continuously decreases as it rotates. A description of each of the parameters appearing in equations (1) – (4) is furnished in Table 1 and a typical generated shell shape is depicted in Fig. 1c and 1d. The shell shape so generated is termed the natural turritella shell.

To minimize the structural parameters, we simplify two of the primary features of the natural turritella shell: the elliptical aperture (Fig. 1d) is assumed to be circular (Fig. 1f), and the plane of the aperture (Fig. 1d) is assumed to be normal to the direction of growth of the whorl. The specific simplifications made on each parameter to generate the simplified turritella are described in the third column of Table 1. Following the incorporation of the above-mentioned simplification, we are left with the following five mathematical parameters for the parametric study: A , β , R , α and θ_0 . The resulting simplified equations are shown in equations (5), (6), and (7), respectively.

$$x = [A \sin \beta \cos \theta + R \cos (s) \cos (\theta)] \cdot e^{\theta \cot \alpha} \quad (5)$$

$$y = [-A \sin \beta \sin \theta - R \cos (s) \sin (\theta)] \cdot e^{\theta \cot \alpha} \quad (6)$$

$$z = [-A \cos \beta + R \sin (s)] \cdot e^{\theta \cot \alpha} \quad (7)$$

where,

$$\theta \in [0, -\theta_0] \text{ and } s \in [-267^\circ, 39^\circ]$$

Table 1

Description of each parameter in the 11-parameter model for molluscan shells and simplification employed for the turritella shell in the present study.

Parameter	Mathematical Description	Interpretation/Simplification for Turritella Shell
θ	Polar coordinate of rotation from positive x-axis (in radians)	Specifies the number of rotations of the aperture that generate the whorls; Range: $[\theta_0, 0]$, $\theta_0 < 0$
α	The equiangular angle of the logarithmic spiral (in degrees)	Specifies the pitch of the whorls
s	Angular parameter generating the elliptical aperture curve	Range: $[-267^\circ, 39^\circ]$; restricted from $[-270^\circ, 90^\circ]$ to avoid redundant internal surfaces
a	Length of the major axis of the initial elliptical aperture	Aperture is assumed circular.
b	Length of the minor axis of the initial elliptical aperture	$a = b = R$
μ	Amount of leaning over of aperture or rotation about y-direction (in radians)	Neglected, = 0
Ω	Amount of azimuthal rotation of aperture or rotation about z-direction (in radians)	Neglected, = 0
ϕ	The tilt of the major axis of the aperture from horizontal xy-plane or rotation about x-direction (in radians)	Redundant since the aperture is assumed circular, = 0
A	Distance between initial aperture centre and Cartesian origin	Specify the location of the centre of the initial aperture; to be determined by root cone parameters H_c and R_c , and R
β	The angle between the shell axis and the line joining the initial aperture centre to Cartesian origin	
D	Sense of coiling of the shell	Assumed dextral, = 1

Since singly varying each of these *mathematical parameters* may distort the shape of the turritella, and varying them directly may not be meaningful for a hydrodynamic study, we map them to five new *auxiliary parameters*, as shown in Table 2, namely $R_c, H_c, \alpha', \delta$ and n . The physical significance of each of the above five auxiliary parameters on the turritella geometry is shown in Fig. 1e and 1f. The mapping is discussed in detail in Appendix A. With these five *auxiliary parameters*, the simplified turritella seashell may be conceptualized as follows: The geometry of the turritella is thought of as an assembly of spiral whorls inside a 'root' cone (outlined faintly in Fig. 1e). This root cone is first specified by the cone height (H_c) and the base radius (R_c). The assembly of whorls begins at the initial (circular) aperture, whose centre lies in the plane of the root cone base. Three parameters need to be specified to generate spiral whorls inside this root cone: i) the pitch angle of the whorls (α'), ii) the depth of the ridges between the whorls, which is specified as a ratio (δ), and iii) the number of rotations of the whorls (counting from the base of the root cone) (n). These five auxiliary parameters of the simplified turritella are classified as root cone parameters and whorl parameters, shown in Table 2. As per this conceptualization, these new *auxiliary parameters*, namely: $R_c, H_c, \alpha', \delta$ and n can

Table 2

Description of five auxiliary parameters of the simplified turritella shell.

Auxiliary Parameter	Description	Type
R_c	Root cone radius	Root cone parameters
H_c	Root cone height	
α'	Pitch Angle of whorls	Whorl parameters
δ	Depth ratio of whorls	
n	Number of whorls	

be singly varied without distorting the shape of the turritella. Hereafter, the *auxiliary parameters* will be referred to as just 'geometric parameters'.

2.2. CFD and mesh independence study

The finite element package COMSOL Multiphysics® v5.5 was used to solve the flow field. The 3D model surfaces of the turritella geometry were created using equations (5) – (7) in Matlab. These surfaces were then processed in Meshlab (Cignoni et al., 2008; Kazhdan and Hoppe, 2013) and imported into COMSOL®. For the simulation studies, the turritella insert was placed at a distance of approximately 5 cm from the beginning of a cylindrical pipe of length 1.5 m. The length of the pipe was chosen to be sufficiently long for mixing to occur (see section 3.1). The *Shear Stress Transport (SST) turbulence model* was used to solve the turbulent flow field around the turritella insert (SST model equations are given in Appendix C and COMSOL physics interface details are given in Appendix D). The SST model has been known to perform well for simulations of V-cone flow meters, whose geometry resembles that of the present study (Sheikh et al., November 2019). The single-phase working fluid was water, assumed to be incompressible and isothermal at room temperature. A fully developed velocity profile was provided at the inlet, and average inlet velocity values were reported.

The fineness of the mesh was selected through a mesh independence study, wherein 3 meshes of increasing fineness were used to solve for the flow field. The details of these meshes are given in Table 3. The volume-averaged velocity and pressure were used to evaluate the error between the three meshes. For subsequent studies, the fineness setting of the second mesh was chosen as a good balance between solution accuracy and computational cost. The pipe domain was divided into two sections for efficient meshing: the insert section (with tetrahedral meshing) and the pipe section (with a sweeping grid mesh).

2.2.1. Mixing quantification

Various methods are available to quantify the extent of mixing in inline flow systems (Aubin et al., 2010). In this work, we have used the universal metric of Shannon information entropy (Shannon et al., 1950; Snell et al., 1959), similar to that employed by Fodor et al. (Fodor et al., 2009) and Zhang et al. (Zhang et al., October 2018) in the context of static mixers. In this study, 3000 tracer particles of negligible diameter (1 μm) and having density (1000 kg/m³) equal to the working fluid (water) are introduced at the inlet using the *Particle Tracing Module* in COMSOL (details of the physics interface are given in Appendix D). Half of these particles are coloured blue, and half are coloured red, divided by a diameter at the inlet cross-section (Fig. 2 left side). Cross-sections at different positions in the aft of the insert are taken for data sampling. As a particle passes through the sampling cross-section, its position is recorded. The recorded positions are then used for further entropy calculation.

The cross-section is divided into 144 bins by polar coordinate lines (18 angular divisions $\times$ 8 radial divisions). For each bin, the fraction of red particles to the total number of particles is calculated as f_i , where, $0 < f_i < 1$. f_i may also be interpreted as the probability of finding a red particle in the i^{th} bin. Thus, the set of f_i is a probability distribution of red particles across the bins of the cross-section. The entropy of the i^{th} bin is given by,

$$s(f_i) = -[(1 - f_i)\log_2(1 - f_i) + f_i\log_2 f_i]. \quad (8)$$

Table 3
Mesh resolutions used in the mesh independence study.

Mesh	Cells across d_p	Total mesh cells	% Error compared to Third Mesh	
			Vol ^m averaged velocity	Vol ^m averaged pressure
First	~22	228,138	1.72 %	3.02 %
Second	~27	518,725	0.92 %	0.35 %
Third	~43	3,987,194	–	–

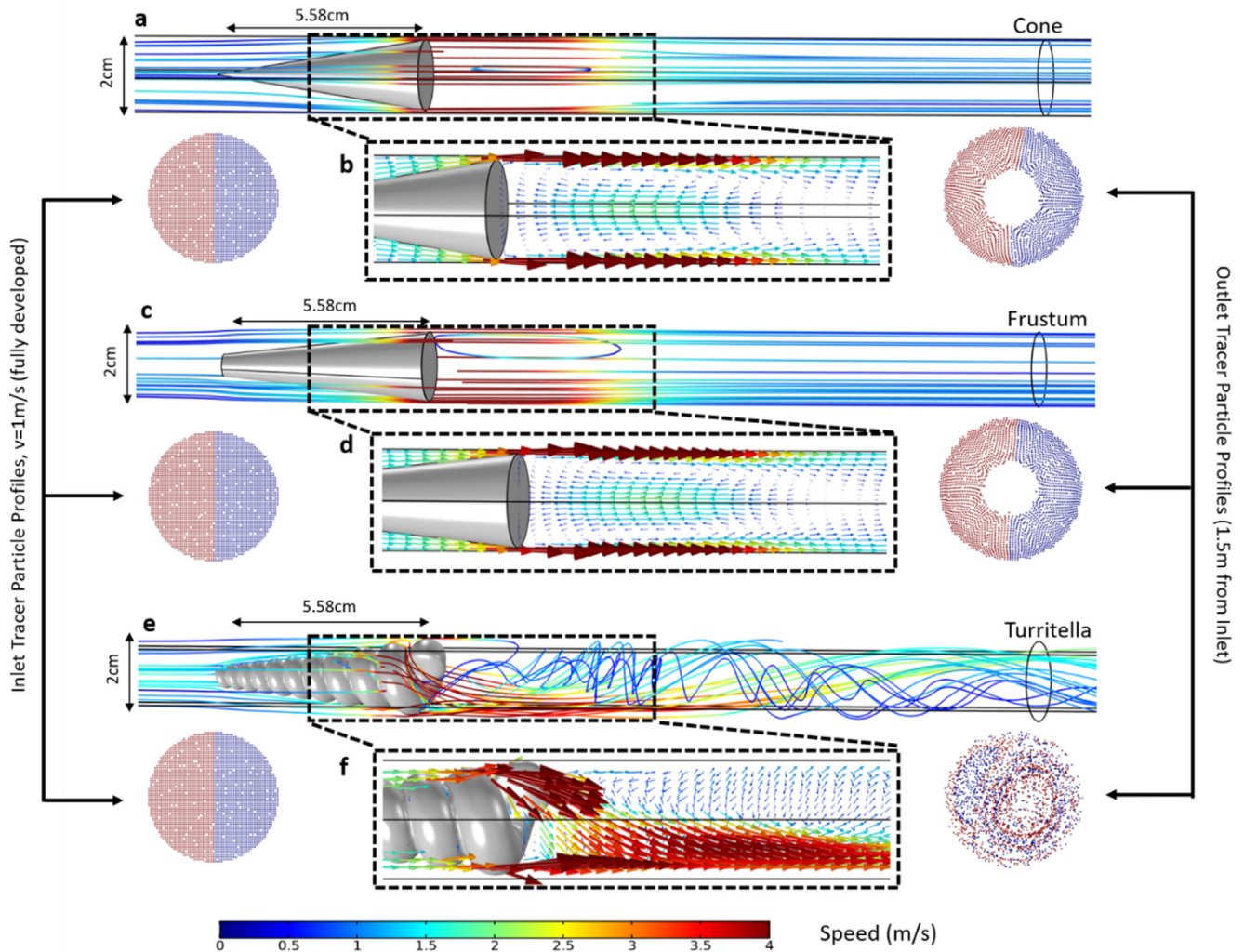


Fig. 2. A section of the pipe shows selected trajectories of tracer particles for 3 different inserts: (a) cone, (b) frustum, and (c) turrutella. The colour denotes the velocity magnitude of the tracer particles. The colour scale is restricted for better visualization. The maximum velocity is $\sim 8\text{ m/s}$. Insets portray a velocity arrow plot for the (d) cone, (e) frustum, and (f) turrutella insert. The arrow length is proportional to the local velocity magnitude.

In a well-mixed cross-section, the probability of finding a red particle (f_i) in any given bin is close to 0.5, thus yielding a maximum entropy of $s(f_i) = 1$ for that bin. In a poorly-mixed cross-section, however, many bins will have only red or only blue particles in them, implying that the probability of finding a red particle (f_i) in such bins is either close to 0 or 1, yielding a minimum entropy of $s(f_i) = 0$. The total entropy of the cross-section is then calculated as an area-weighted average over all the bins, mathematically defined as:

$$S = \frac{1}{A} \sum_i s(f_i) A_i \quad (9)$$

where, A_i is the area of the i^{th} bin and A is the sum of the area of all the bins, i.e., the area of the entire cross-section.

3. Results and discussions

3.1. Demonstration of inline mixing introduced by the turrutella insert

To establish that a turrutella insert can induce mixing in a pipe, we compare the velocity flow field and tracer particle trajectories around the turrutella insert alongside two similarly sized geometric inserts: a cone and a frustum. All the inserts are of comparable dimensions (outlined in Fig. 2); the lengths of all the inserts are kept equal (5.58 cm), the breadths of the apex are equal for the frustum and turrutella (0.30 cm), and the breadth of the broader ends of the frustum and cone (1.844 cm) are kept such that the cross-sectional areas in both cases are equal. Operating conditions like average inlet velocity ($v = 1\text{ m/s}$) (Fig. 2, left side), pipe

diameter ($d_p = 2$ cm), pipe length (1.5 m) and no-slip conditions were kept invariant for all three inserts. All inserts are kept at a distance of ~ 5 cm from the inlet. The turrutella geometry was prepared as described in the methods section, with the parameters as specified for model number P-1 in Table B.2 in Appendix B. The turrutella insert is placed in accordance with the most stable apex-up-current configuration of the turrutella seashell observed on the beach. Tracer particles are provided at the inlet and are coloured red and blue for visualization and quantification of the mixing (Fig. 2, left side).

It is immediately observed from the tracer particle trajectories in Fig. 2a, 2c, and 2e that swirling occurs only in the case of the turrutella insert, while in the case of the cone and frustum the particle trajectories follow a straight line. Additionally, the pressure drop across the pipe for the turrutella insert is remarkably less than that for the other inserts, $\Delta p_{\text{turrutella}} = 9.3 \text{ kPa} < \Delta p_{\text{cone}} = 14.1 \text{ kPa}$, $\Delta p_{\text{frustum}} = 14.4 \text{ kPa}$. The broader end of the insert is where the fluid experiences the highest constriction. In the case of the frustum and the cone, the fluid flowing out of this constricted region primarily has an axial velocity component with negligible transverse velocity. This can also be seen from the velocity arrow plot (Fig. 2b and 2d) where the velocity arrows are directed parallel to the axis of the pipe for the cone and frustum while emerging from the constricted region between the pipe and the insert. The low-pressure wake formed in the immediate aft of these geometric inserts pulls this flow inwards, causing a backflow. This can be seen in the insets of Fig. 2b and 2d (central part of the cross-section of the pipe), where the velocity arrows point towards the insert and against the general forward flow direction. Thus, the axial component of the flow is sacrificed, which leads to a higher pressure drop across the whole unit. In the case of the turrutella insert, however, the fluid is guided by the spiral whorls of the insert in a more streamlined fashion. When the fluid emerges from the constricted region, it has both an axial component as well as a transverse velocity component. This can be seen from the velocity arrow plot (Fig. 2f) where the velocity arrows are at an angle with respect to the pipe axis for the turrutella seashell after emerging from the constricted region between the insert and the pipe. The low-pressure wake pulls this flow inwards, like in the cases above, but here, the transverse component of the flow is sacrificed instead of the axial component, greatly reducing the amount of backflow. This can be seen in Fig. 2f, where the velocity arrows pointing in the reverse direction after the turrutella shell insert are of less magnitude. This in turn, reduces the pressure drop across the pipe. Further, the 'pull' from the low-pressure zone of the wake induces a tangential motion in the flow and the fluid begins to swirl leeward as seen in the particle trajectories. In contrast, no swirling is present in the case of the geometric inserts (cone and frustum). Quantitatively, the average transverse velocity (averaged over the volume of the pipe) is larger for the turrutella insert as compared to the other inserts, $\bar{v}_{\perp, \text{turrutella}} = 0.134 \text{ m/s} > \bar{v}_{\perp, \text{cone}} = 0.007 \text{ m/s}$, $\bar{v}_{\perp, \text{frustum}} = 0.007 \text{ m/s}$, where $\bar{v}_{\perp} = (\bar{v}_x^2 + \bar{v}_y^2)^{1/2}$.

The tracer particle outlet profiles (Fig. 2, right side) show that the red and blue tracer particles are well mixed only in the case of the turrutella insert, indicating mixing in the fluid. Conversely, for the cone and the frustum inserts, the red and blue tracer particles remain segregated. To establish a quantitative comparison of the extent of mixing, the Shannon entropy was evaluated for the tracer particle distribution at the exit for each of the inserts as discussed in Section 2.2.1. The entropy was found to be ~ 0.34 for the turrutella insert. This is much higher than the mixing entropy of 0.03 for the cone and frustum inserts. The above results conclusively demonstrate that the turrutella insert, when placed in a pipe, induces mixing and can potentially function as a static mixer. The above discussion and the comparison with different geometric

inserts indicate that the mixing behaviour is an outcome of the unique geometry of the turrutella insert constituting growing spiral whorls.

The mixing observed in the turrutella geometry does not occur immediately after the insert, rather it evolves along the length of the pipe following the turrutella insert. Fig. 3 shows the evolution of the tracer particle distribution at selected cross-sections along a 3 m pipe length (~ 58 turrutella insert lengths). After passing the turrutella insert, the mixing entropy corresponding to the tracer particle distribution increases till a length of approximately 0.5 m (~ 10 insert lengths), after which it saturates at a value of ~ 0.34 (Fig. 3a). Visually, after a pipe length of 1.5 m (~ 29 insert lengths), the cross-sectional distribution of the tracer particles remains nearly invariant (Fig. 3b), indicating that the swirling motion of the fluid dampens. Thus, it is reasonable to restrict the pipe length to 1.5 m. Having established the mixing capability of the turrutella insert, we proceed to explore the mechanism and characteristics of this turrutella-inspired static mixer configuration.

3.2. Mechanism of mixing

The mixing process can be described in terms of a two-step mechanism: *flow splitting due to asymmetric constriction* followed by *shearing due to the pipe wall*. In the following section, we discuss this two-step mechanism and support the proposed theory with parametric studies on operating parameters like pipe diameter (d_p), average inlet velocity (v), and slip conditions of the geometry surface.

3.2.1. Flow splitting due to asymmetric constriction

The spiral geometry of the turrutella insert is inherently asymmetric and leads to asymmetric constriction of the fluid flow at its broader end. As a result, the incoming flow is split into two streams, as shown in Fig. 4a. To understand the dynamics of this asymmetric constriction, we define a cross-section near the broader end of the insert (approximately positioned such that it passes through the centre point of the circular aperture); and we divide the cross-section into two zones: Zone A which comprises of the apertural gap, and Zone B which comprises of the remaining semi-annular gap between the pipe wall and the insert, see Fig. 4b. The gap that exists between the pipe wall and where the whorls terminate at the aperture is termed the 'apertural gap'. The fluid flowing through Zone A is referred to as stream A, while that flowing through Zone B is referred to as stream B. In the absence of any strict boundary between the apertural gap and the annular gap, we have used a quadrant-based division to demarcate the two zones for quantification. The division is made by specifying constants c_1 and c_2 such that all points (x, y) in the cross-sectional plane satisfying $x < c_1$ and $y < c_2$ constitute Zone A, while its complement forms Zone B, as shown in Fig. 4b.

The spiral whorls of the turrutella help in guiding a fraction of the fluid into zone A, while the rest of the fluid flows through zone B. Between stream A and stream B thus formed, stream A has a transverse component of velocity, unlike stream B, see Fig. 4a. The development of this transverse velocity can be attributed to the successively growing spiral whorls, which gradually accelerate the fluid in the transverse direction (Fig. 4c). As an incoming fluid parcel is obstructed by a whorl of the turrutella insert, it experiences two forces exerted by a no-slip insert surface, namely, the reaction force normal to the local surface (N) and the frictional force parallel to the local surface (f), see Fig. 4e. If we resolve N and f into their respective lateral ($N_{\parallel}, f_{\parallel}$) and transverse components ($N_{\perp}, f_{\perp}$), we see that $N_{\perp}$ and $f_{\perp}$ are both responsible for constraining the fluid parcel's velocity in the transverse direction. After emerging from the apertural gap, the whorl-induced trans-

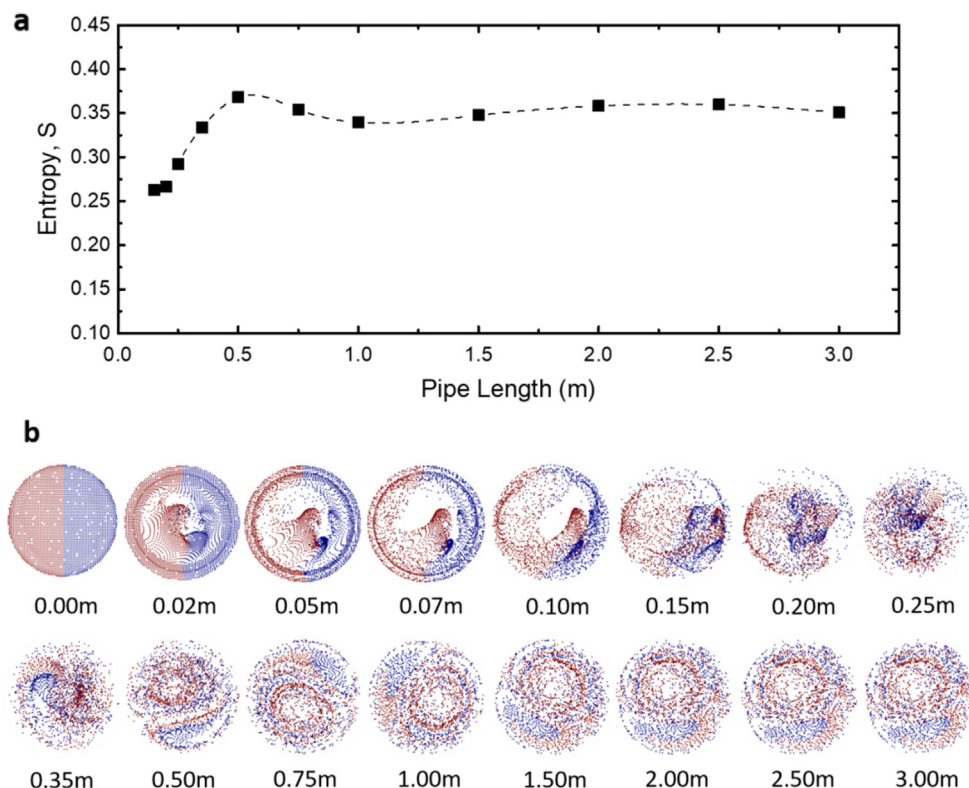


Fig. 3. (a) The value of Shannon mixing entropy plotted along the length of the pipe. (b) Cross-sections sampled at various lengths along the pipe depict tracer particle positions as they pass.

verse component of stream A is further enhanced by the low-pressure wake in the immediate aft (Fig. 4d), which ‘pulls’ stream A inwards. Moreover, stream A, by and large, has a larger velocity than stream B, as is observed from the colour scale of the tracer particle velocities in Fig. 4a. This combination of a high-velocity and transverse component of stream A ‘overwhelms’ the relatively low-velocity stream B with a negligible transverse component. This causes a swirling motion leeward of the insert, which forms the first step toward mixing. The swirling continues until it eventually dampens out and the flow regains a developed profile.

The above mechanistic behaviour can be interpreted quantitatively based on the energy carried by streams A and B, respectively. Specifically, in order for stream A to overwhelm stream B, we hypothesize that the kinetic energy per unit length carried by stream A should be at least greater than or equal to that carried by stream B. To this end, we perform a quantitative analysis of the two streams by comparing the integral of the kinetic energy per unit volume evaluated over the respective areas of Zone A and Zone B. This quantity, which may also be interpreted as the kinetic energy per unit length of the stream, is represented by K_i (unit: J/m) and is given by equation (11):

$$K_i = \int_{\text{Zone } i} \frac{\rho |\mathbf{v}|^2}{2} dA, \text{ where } i = A, B. \quad (11)$$

The ratio of K_A/K_B (k ratio) should be sufficiently high so as to ensure stream A ‘overwhelms’ stream B, resulting in sufficient swirling and mixing. As can be seen in equation (11), K_i depends on two controllable parameters: cross-sectional area and fluid velocity. Therefore, the effect of varying the pipe diameter (affecting the cross-sectional area) and the effect of varying the inlet velocity on the mixing performance is studied in the following two subsections to support the proposed mechanism.

3.2.1.1. Pipe diameter study. The diameter of the pipe in which the turrilella insert is placed determines the size of the cross-sectional area through which the fluid flows and, therefore, the area corresponding to the apertural gap and the semi-annular gap. Therefore, varying the pipe diameter is expected to affect the k ratio. This point is reinforced through a study of varying the pipe diameter (d_p) for the same turrilella insert and constant average inlet velocity of 1 m/s (parameter specifications given in Table B.2 in Appendix B).

As shown in Fig. 5a, there is a rapid decline in both mixing entropy (S) and pressure drop (Δp) on increasing the pipe diameter. There is over a 90 % decrease in pressure drop (from 18 kPa to 1.5 kPa) as the diameter is increased by only 40 % (from 2 cm to 2.8 cm, i.e., from Re 20000 to 28000), indicating the advent of less constriction between the largest whorl and the wall. A plot of the k ratio reveals that as the pipe diameter is increased, the k ratio falls (Fig. 5b). The less constricted environment at larger pipe diameters allows for more fluid to flow through Zone B, resulting in a higher K_B value than K_A , yielding a lower k ratio. As a result, the relatively low-energy stream A which carries a transverse component of momentum, is ‘dwarfed’ by the momentum of high-energy stream B and is unable to induce sufficient swirling motion. Insufficient swirling consequently leads to insufficient mixing, which is apparent from the outlet tracer particle profiles (Fig. 5a) and the low value of mixing entropy 0.12 for $d_p = 2.8$ cm. At lower pipe diameters, however, the increased constriction causes more fluid to flow through Zone A, thus yielding a higher k ratio (Fig. 5b). For the smallest pipe diameter case with $d_p = 2$ cm, a high k ratio of 1.33 is observed. With increasing k ratios, the extent of swirling increases in the flow. The result is good mixing at the outlet, as is evident from the outlet tracer particle profiles (Fig. 5a). Thus, the above results are in line with the proposed mechanism that a sufficiently high k ratio is the requirement to realize adequate mixing.

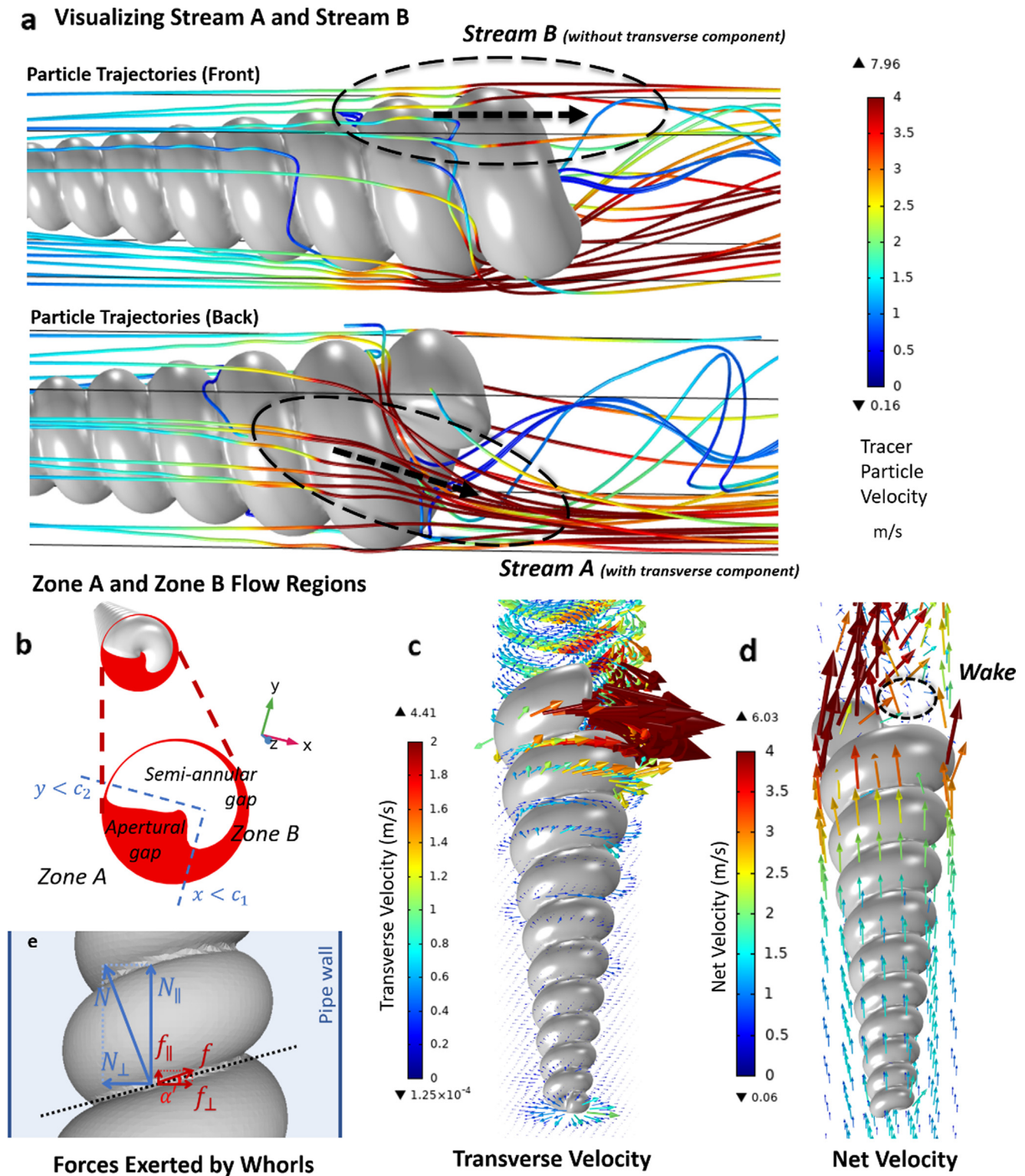


Fig. 4. (a) Selected particle trajectories depicted from the front and back depicting stream B and stream A respectively. (b) The quadrant-based division used to demarcate Zone A and Zone B. (c) Plot of transverse velocity arrows along numerous cross-sections along the pipe. The arrow length is proportional to the local velocity magnitude. Both colour scales are restricted for better visualization. Maximum and minimum values of velocity are mentioned above and below the scales. (d) Plot of velocity arrows and the depiction of the wake region. (e) A schematic diagram of the lateral and transverse components of frictional (f) and normal (N) forces exerted by the turrilella surface on an incoming fluid parcel.

3.2.1.2. Velocity study. In addition to the cross-sectional area through which the fluid flows, the fluid velocity dictates the kinetic energy per unit length K_i , for each of Zone A and B. Therefore, the effect of varying the inlet velocity (0.4–3 m/s) for the same turrilella insert

placed in two separate pipes of different pipe diameters ($d_p = 2$ cm and 2.2 cm) on the mixing entropy has been demonstrated here, and the inferences are correlated to the proposed mechanism. The parameter specifications are given in Table B.1.

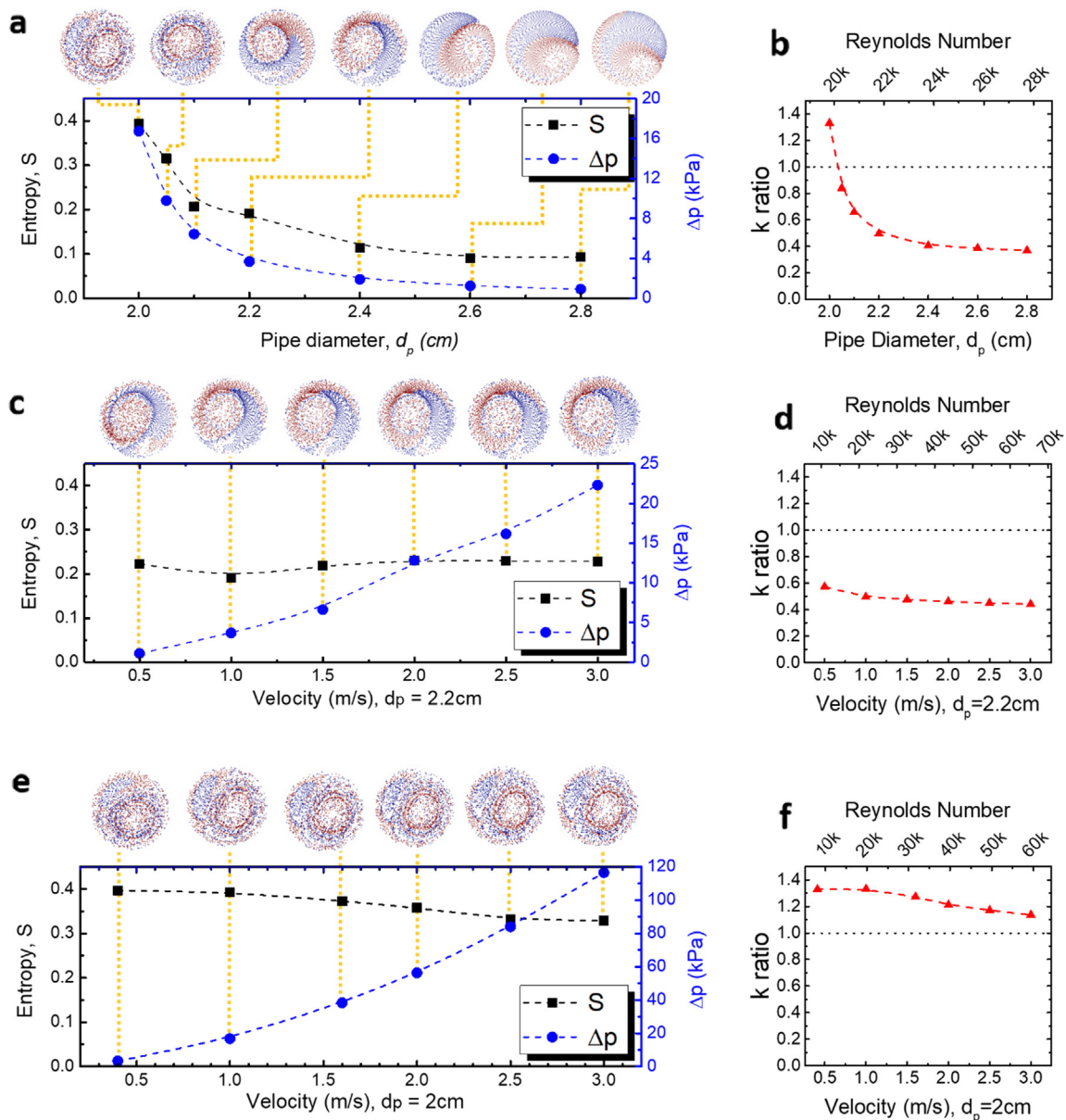


Fig. 5. The mixing entropy and pressure drop characteristics for (a) the pipe diameter study, (c) the inlet velocity study with $d_p = 2.2$ cm and (e) the inlet velocity study with $d_p = 2$ cm. The corresponding outlet tracer particle profiles are shown above each plot. (b,d,f) The corresponding ratios of the kinetic energy per unit length of stream A to that of stream B, i.e., K_A/K_B , are plotted, along with information about the Reynolds number. A horizontal dotted line at the value of 1 is shown for reference.

In both pipes, the mixing entropy and pressure drop each exhibit the same trend: the mixing entropy remains within 0.07 of its value, while the pressure drop increases with increasing velocity (Fig. 5c and 5e). The near-constant mixing is apparent from the visually similar outlet profiles of the tracer particle distribution. The value at which the mixing entropy is constant, however, is higher for the more constricted geometry ($S \sim 0.39$ for $d_p = 2$ cm), than that of the less constricted geometry ($S \sim 0.22$ for $d_p = 2.2$ cm). This is attributed to their respective k ratios. In the $d_p = 2.2$ cm case, the k ratio remains low ($\lesssim 1$) and within a narrow range of 0.48 ± 0.05 (Fig. 5d); whereas in the $d_p = 2$ cm case, the ratio is high ($\gtrsim 1$) and within a narrow range of 1.24 ± 0.08 (Fig. 5f). Thus, we see that the near-constant mixing entropy in both cases correlates well with the near-constant k ratio of the split streams. We also note that the poor mixing observed in a large diameter pipe cannot be compensated for by providing a larger inlet velocity. Instead, the entropy remains constant and does not depend on the inlet velocity in this range. Increasing the flow velocity essentially increases

the kinetic energy of the individual streams, however, that *does not* translate into a higher k ratio and higher mixing. This emphasizes the fact that the *ratio of kinetic energies* K_A/K_B is the controlling factor for the extent of mixing and not just the individual magnitudes of K_A and K_B . The above behaviour can be perceived as a near-constant entropy-velocity characteristic and may have a promising potential application in an industrial setting.

3.2.2. Shearing due to pipe wall

The second part of the mechanism consists of the shearing effect that follows the constriction effect in the aft of the turrillita insert. As described in Section 3.1, the mixing entropy of the tracer particles evolves along the pipe length following the turrillita insert. In this section, we show that, in addition to the high k ratio, the shearing effect due to the no-slip condition at the pipe wall is crucial in sustaining swirling for a long length afterward, resulting in the mixing.

To understand the significance of this pipe shear, we first demonstrate a couple of simulations in which the usual no-slip (NS) boundary condition at the pipe wall (Fig. 6a) is replaced with a complete slip (CS) boundary condition (Fig. 6b). All other operating parameters and turrutella insert geometry are kept invariant. Arrow plots of transverse velocity are shown at four different cross-sections in the aft of the turrutella insert, each spaced 4 cm apart. Selected tracer particle trajectories and outlet profiles are also shown. It is interestingly seen that the outlet tracer particle profiles are well mixed and hence the mixing takes place only when there is no slip at the pipe wall, even though the k ratio is equally high in both cases: 1.33 for wall = NS and 1.29 for wall = CS. The particle trajectories also indicate a drastic reduction in swirling motion on the removal of the wall's no-slip boundary condition. Thus, we infer that the boundary layer at the no-slip wall assists in creating a high shear environment in the fluid leeward of the insert. This sustained shear is responsible for good mixing and visually manifests as swirling motion of the particle trajectories.

In the next couple of simulations, we investigate what occurs when the boundary condition of the insert is changed (Fig. 6c and

6d). Since the interaction of the fluid with the insert is for a shorter duration than that with the pipe wall, it is expected that the insert boundary condition will have a less prominent effect on the flow. As the no-slip condition at the insert is changed in both wall = NS (Fig. 6c) and wall = CS (Fig. 6d) cases, no major difference is observed in terms of mixing. However, a minor difference is observed in terms of the transverse velocity of stream A, which is greater when insert = NS in both cases. Between Fig. 6a and 6c, this difference in velocity is apparent by comparing the transverse velocity magnitudes at the second cross-section from the insert. As a result, the insert/wall = NS/NS case presents itself with a slightly higher mixing entropy ($S_{NS/NS} = 0.39$) as compared to the insert/wall = CS/NS case ($S_{CS/NS} = 0.33$). Between Fig. 6b and 6d, a higher transverse velocity is again observed for the insert = NS case, which is apparent at the first cross-section itself. However, in both cases, the mixing is negligible.

In conclusion, by switching the boundary conditions at the insert and the wall each, we demonstrated that shearing due to the pipe wall plays a crucial role in the mixing mechanism. The boundary condition at the insert, however, does not play a major role in affecting the mixing. Nevertheless, the no-slip condition

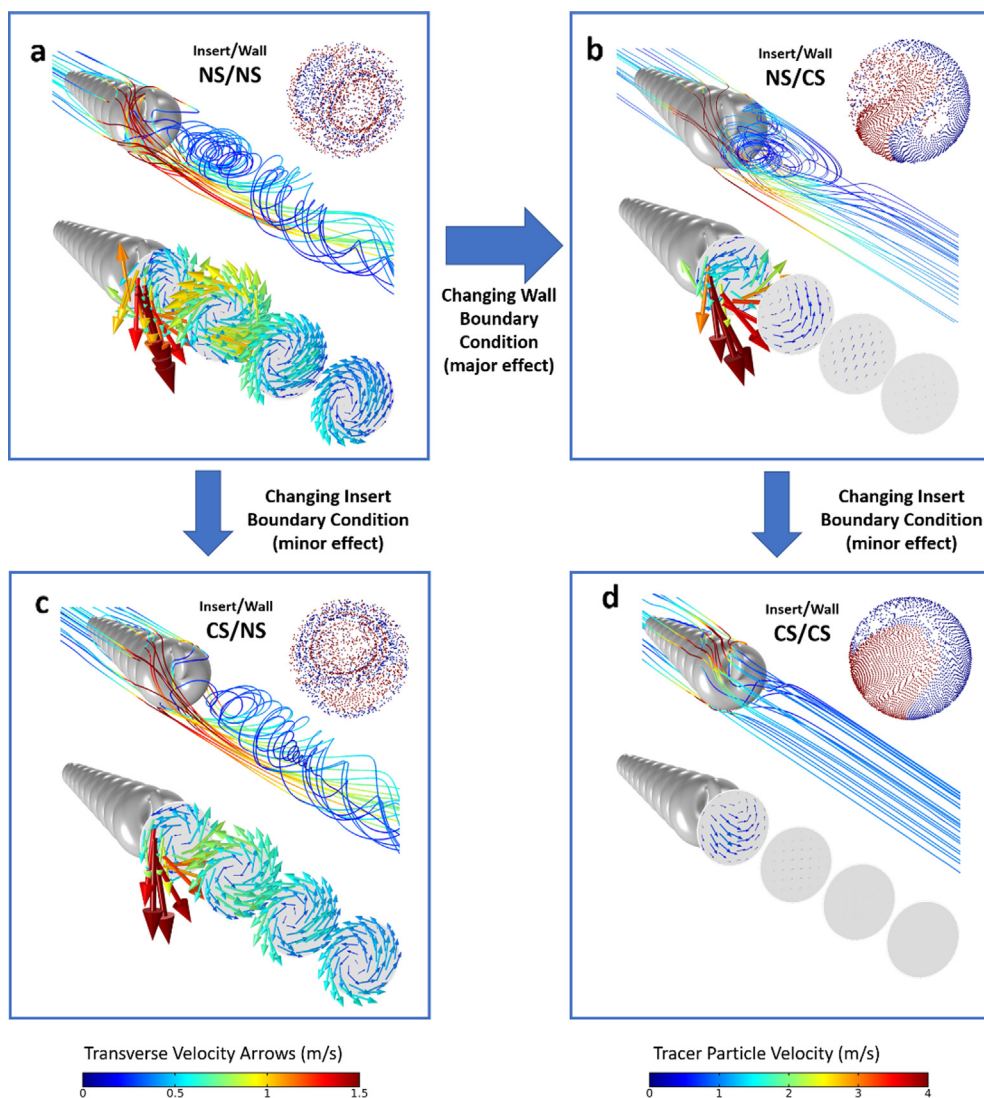


Fig. 6. Depiction of the outlet tracer particle profiles, selected tracer particle trajectories, and transverse velocity plots on 4 successive cut planes in the aft of the turrutella insert (spaced 4 cm apart) for models having different slip conditions at the insert and wall surfaces. The arrow length is proportional to the local velocity magnitude. The colour scale is restricted for better visualization.

would be preferred for the insert to allow for a greater transverse velocity of stream A.

Thus, through these three studies of pipe diameter, inlet velocity, and boundary conditions, we have elucidated a two-step mechanism of mixing for the turrutella static mixer. The asymmetric constriction created by the turrutella inserts causes flow-splitting into two streams, A and B. The whorls of the turrutella insert impart the transverse velocity to stream A emerging from the apertural gap. A high k ratio of the streams coupled with shearing due to the pipe wall forms a favourable premise for mixing to occur in the aft.

3.3. Effect of the turrutella geometry on mixer performance

Having understood the mechanism of mixing, we further investigate whether or not the geometric parameters of the turrutella insert have any effect on the mixing entropy or the pressure drop characteristics. For this study, we limit the variation in the geometric parameters such that the overall shape resembling that of the turrutella seashell is retained. As discussed in Section 2.1, we develop a simplified 5-parameter model that captures the most salient features of the natural shell. The geometry of the simplified turrutella is conceptualized as an assembly of spiral whorls inside a root cone. Thus, the root cone with parameters H_c and R_c , and the whorls with parameters α' , δ , and n are used to describe the turrutella insert geometry (see Table 2). Fig. 7 illustrates the variation in the turrutella geometry when each of these geometric parameters (except R_c) is varied independently. The resultant outlet profiles and entropy and pressure drop characteristics are depicted alongside. The operational parameters (d_p and v) and R_c are kept constant while varying the geometry to ensure a similar constriction area of the incoming stream. The values of the parameters used in simulating each model are given in Table B.1. in Appendix B.

It is notable that, in each case, the mixing entropy is relatively a weak function of the geometric parameters; the value of S does not change by more than 0.05 (characteristic plots shown in Fig. 7a–d). This is also observed from the outlet profiles, which appear to be similarly mixed in each case. Stream splitting occurs equally well in each geometry. Thus, for practical purposes, we regard the mixing entropy to be independent of the geometric parameters.

On the other hand, the geometric parameters have a considerable effect on the pressure drop across the unit. Although stream splitting is ubiquitous among the different models, some geometries are more favourable for streamlined splitting than others. This is reflected through the pressure drop characteristics. For turrutella inserts with varying pitch angle α' (Fig. 7a), the pressure drop is seen to decrease (~ 4 kPa) with increasing pitch angle. As the pitch angle increases, the spiral whorls are more inclined from the cross-sectional plane. Therefore, the magnitude of the lateral $N_{||}$ force exerted by the whorls (Fig. 4e) on the fluid decreases, which leads to less form drag. The depth ratio δ has a minute impact on the pressure drop: a ~ 1 kPa increment is observed at a high depth ratio (Fig. 7b). Deeper ridges provide more scope for the retardation of incoming fluid parcels. The root cone height H_c – which controls the slenderness or stoutness of the insert – has a negligible impact on the pressure drop, i.e., within ~ 0.5 kPa (Fig. 7c). The pressure drop characteristic exhibited by the number of whorls n shows a non-monotonic trend (Fig. 7d). This is due to the competing effects of the decreasing resemblance of the insert to a bluff body and the increasing number of whorls that exert $N_{||}$ force on the incoming fluid parcels as n is increased.

In conclusion, the geometric parameters do not significantly impact the mixing entropy, however, they can be fine-tuned to reduce the pressure drop across the pipe. Among the turrutella

inserts studied, the reduction in pressure drop between the insert having the highest Δp (N-3) and the insert having the lowest Δp (A-3) is 43 %. The ramifications of this behaviour will be discussed in the following section.

3.4. Key insights obtained from the turrutella-like static mixer

The present study on the turrutella-like geometry as a static mixer may provide key insights into the design of a new class of static mixers. These insights may prove useful both to extend research on turrutella-like static mixer geometries and, to improve studies on general static mixers which may not be turrutella-shaped.

3.4.1. Insights on the design of general static mixers

General static mixers may be designed to emulate the velocity-invariant near-constant mixing characteristic of the turrutella-like static mixer as described in Section 3.2.1.2. This characteristic is a consequence of the asymmetric flow split, which creates a high-velocity transverse stream and a low-velocity axial stream. The ratio of the kinetic energy between the two streams governs the extent of mixing and is only dependent on the flow constriction created due to the geometry and does not depend on the flow rate. It may be possible to engineer geometries of static mixers which operate on this asymmetric flow-splitting concept. The consequent near-constant mixing output is veritably desirable in industrial processes that often operate at a pseudo-steady state, where constant mixing is prescribed at the outlet despite fluctuations in the inlet flow rate.

The turrutella-like static mixer also underscores the importance of leveraging the slip or no-slip boundary conditions at the geometric surfaces of a static mixer, as demonstrated in Section 3.2.2. Typically, in inline flow systems, the no-slip boundary condition leads to major frictional losses over long distances. These frictional losses are ubiquitous in any process industry and are often perceived as an unavoidable loss of energy. However, the turrutella-like insert shows that this 'lost energy' in the form of pipe shear may resourcefully be used to one's advantage to induce mixing. In many static mixer case studies, the role of the slip/no-slip boundary condition at various internal surfaces is rarely addressed. With the advent of affordable superhydrophobic coating technologies (Sow et al., 2021; Lee et al., 2016), such studies and manufacturing techniques are becoming increasingly viable. The drastically negligible amount of mixing observed in the case where pipe shear was absent (Fig. 6a and 6b) emphasizes just how important boundary conditions are in an inline flow system. Thus, it will be worthwhile to investigate the role of slip/no-slip boundary conditions in existing static mixers and in future static mixer designs.

3.4.2. Insights specific to turrutella-like static mixer geometries

Coming to insights more specific to turrutella-like static mixer geometries, in Section 3.3, we demonstrated that it is possible to capitalize on minimizing pressure losses by fine-tuning the geometry of the turrutella insert. Among the different inserts studied, the reduction in pressure drop between the insert having the highest Δp (N-3) and that having the lowest (A-3) was over 40 %. In process industries with continuous flow systems, such a reduction in pressure drop can greatly reduce long-term operating costs. This indicates a promising potential for further optimization studies of turrutella-like threaded inserts. While results here were demonstrated for a simplified 5-parameter model, future work may employ the more elaborate 11-parameter model, allowing greater control over the insert geometry.

In terms of a manufacturing perspective, the turrutella-like geometry—although seemingly complex—can be manufactured by a variety of available methods. Subtractive manufacturing tech-

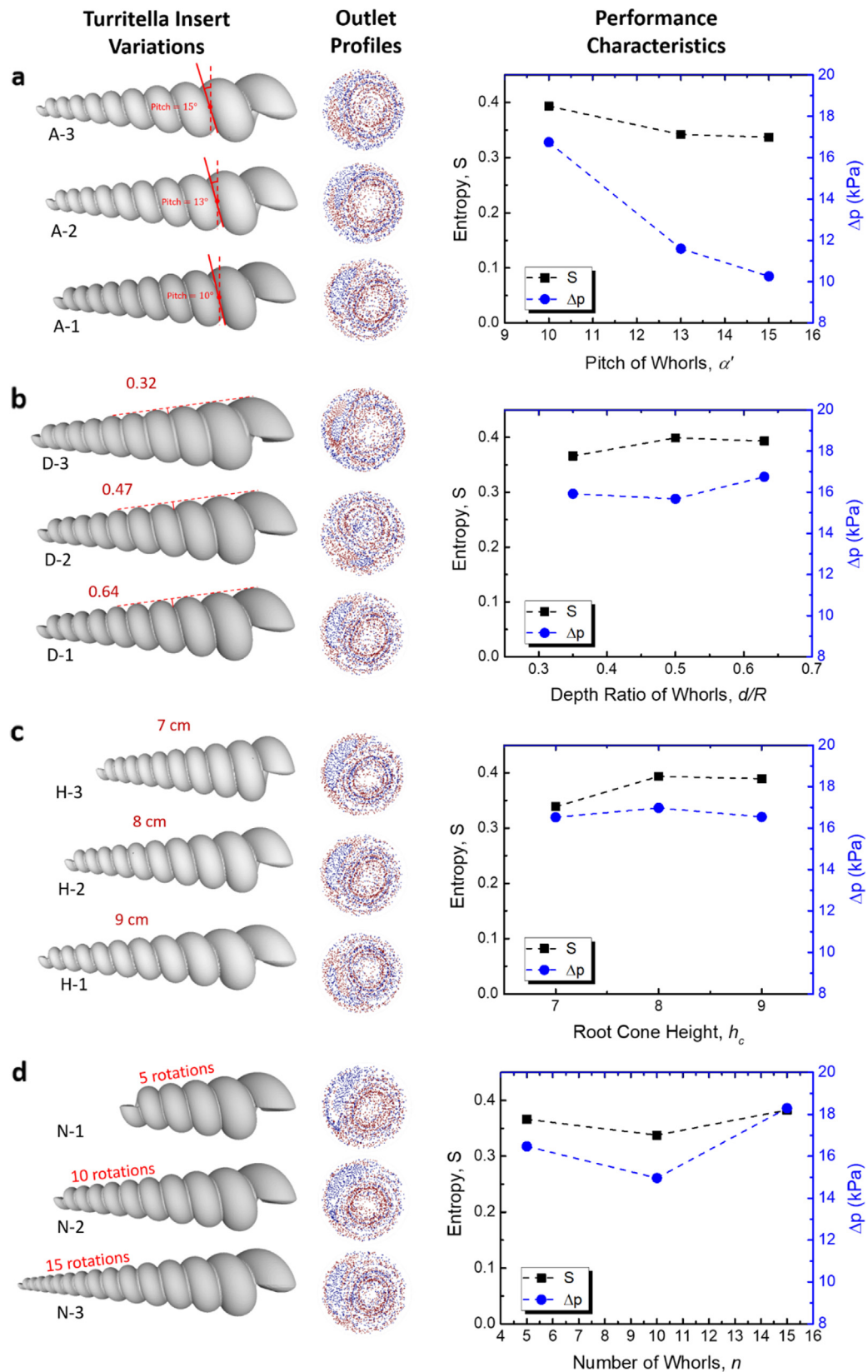


Fig. 7. Geometry of the turrutella-like inserts, resultant outlet profiles and entropy and pressure drop characteristics for each geometric parameter: (a) pitch angle of whorls (α), (b) depth ratio of whorls (δ), (c) root cone height (H_c) and (d) number of whorls (n) respectively.

niques such as traditional mill manufacturing can be employed to machine spiraling channels along the length. Mass production can also be achieved through plastic moulding. Additionally, with the

advent of 3D printing technologies, complex geometries can be 3D-printed with relative ease. Further, 3D printing can offer precise parametric control over the geometry of the insert during

experimentation since the geometry is directly generated from mathematical equations. Regarding the positioning of the insert in the pipe, a single tie rod may be used to pivot the insert at the center of the pipe. The tie rod can be positioned near the apical region of the insert, where it is least likely to affect the flow field. The natural ability of the turrítella to orient parallel to the flow can ensure its correct positioning during operation.

Furthermore, the turrítella insert, like any other insert-type static mixer, is modular in nature. This automatically augments its utility in MAGIC (Modular, AGile, Intensified, and Continuous) processes (Ranade et al., 2015), which are becoming popular in modern specialty chemical plants and process engineering applications that are moving away from traditional batch reactor methods. Since specialty chemical plants often work with non-Newtonian fluids, it may be worthwhile to investigate the mixing behaviour of the modular turrítella-like static mixer in such scenarios. Modularity also enables the usage of multiple inserts in succession to enhance mixing and transport rates according to the application requirement. To characterize these transport rates, in addition to the universal Shannon mixing entropy metric, it may be useful to include different measures of mixing computed from the standard deviation of passive tracer concentration fields (Mariotti et al., 2019).

Finally, the hypothesis upon which this study is based may be of interest to zoologists and marine biologists in understanding how invertebrates like the turrítella cope with flow through morphological adaptations over the course of evolution (Statzner, 2008).

4. Conclusions

In this study, we began from beach observations and hypothesized that the eccentric burrowing behaviour of the turrítella might be a corollary of the constriction and shear environment created as a result of the spiral geometry interacting with the sandy surface. Developing upon this hypothesis, we created an artificial environment to enhance this constriction and shear on all sides by placing a simplified turrítella inside a pipe. To do so, a method was developed to create a simplified turrítella insert, whose geometry can be parametrically controlled. We then investigated the flow physics of this geometry and proposed a two-step mechanism of mixing: flow splitting due to asymmetric constriction followed by shearing due to the pipe wall. Mixing is created as a result of pipe shearing, provided that the kinetic energy ratio of stream A to stream B is sufficiently large. Operating parameters like the pipe diameter, inlet velocity, and boundary conditions on internal surfaces were varied to elucidate the mechanism. Higher mixing is observed in lower pipe diameter configurations. Mixing is nearly constant over a wide range of inlet velocities in the flow regime studied. The no-slip boundary condition at the pipe wall is crucial in creating mixing. The geometric parameters of the insert have no significant effect on the mixing entropy; however, they can be tuned to minimize the pressure drop by over 40 %. Finally, we briefly summarized the insights obtained from the mechanism of the turrítella-like static mixer and methods regarding its manufacturing and implementation. Future directions for the design of both general (non-turrítella-like) static mixers and turrítella-like static mixers were discussed.

CRedit authorship contribution statement

Aditya R. Pujari: Conceptualization, Methodology, Formal analysis, Investigation, Visualization. **Sarvgya Kumar:** Conceptualization, Methodology, Formal analysis, Investigation. **Pradeep Kumar Sow:** Conceptualization, Methodology, Supervision.

Data availability

Data will be made available on request.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A

Mapping between five mathematical parameters and five auxiliary parameters

Given a set of five *auxiliary parameters* ($R_c, H_c, \alpha', \delta$ and n), a unique corresponding set of five *mathematical parameters* (A, β, α, R and θ_0) can be obtained. The *mathematical parameters* thus obtained are plugged back into equations (5) – (7) to generate the required turrítella geometry. We briefly describe how the *mathematical parameters* and the *auxiliary parameters* are mapped to each other.

H_c and R_c : The root cone is first defined by its height H_c and base radius R_c (Fig. 1e). Parameters A and β are then obtained by fixing the value of the radius of the circular aperture, R , and employing basic trigonometry. Note that the breadth of the shell is less than the diameter of the base of the root cone.

α : The angle of spiral α which refers to the equiangular angle of the logarithmic spiral, is correlated with the parameter α' , i.e., the pitch angle. The pitch angle is defined as the angle between a tangent to a ridge line and the x - y plane, see Fig. 1e. Smaller values of α correspond to larger values of α' . α' is measured post-construction of the shell. The parameter is carefully varied within the limited permissible range: if α' is increased beyond 15° , the resultant shape will begin to resemble a corkscrew rather than a turrítella.

δ ratio: The depth ratio of the whorls, defined as $\delta = d/R$, is also measured post-construction of the shell. d represents the depth between two whorls and R represents the radius of the largest neighbouring whorl, where both d and R are in the same plane. Due to the self-similar nature of the turrítella, the δ ratio does not depend on the location of measurement of d and R . Hence, for convenience, we take measurements at the initial aperture as shown in Fig. 1f. Wherever the δ ratio is required to remain constant, it was kept within a standard deviation of 0.02.

n : θ_0 which describes the number of radians traversed by the aperture to generate the whorls is restricted to multiples of 2π . Radians traversed are shown in Fig. 1c. We denote this whole number of whorls by the discrete parameter, number of whorls $n = \theta_0/2\pi$. n is counted from the aperture (counts shown in Fig. 1e) and not the apex—contrary to what is done in conchology—to ensure that models with different n still pertain to the same root cone. We restrict $n \geq 5$ heuristically.

Appendix B

Simulation parameter values

Tables B1 and B2.

Table B1

Values of parameters used in each model in the operating parameter studies. All linear parameters (H_c , R_c , R , A , d , d_p) are expressed in centimeters, angular parameters α , ω and β are expressed in degrees, θ_0 is expressed in radians and n is a number. Inlet velocity v is expressed in m/s. δ , wherever constant, is expressed as a mean.

Mod. No.	Control Parameter	Mathematical Parameters						Auxiliary Parameters					Operating Parameters	
		R	A	β	d	α	θ_0	H_c	R_c	δ	ω	n	d_p	v
A-1	Pitch angle of Whorls (α')	0.5	8.02	3.58	0.32	88.9	-20π	8	1	0.62	10	10	2.0	1.0
A-2		0.55	8.01	3.22	0.34	88.8					13			
A-3		0.6	8.01	2.86	0.37	88.7					15			
D-1	Depth Ratio of Whorls (δ)	0.5	8.02	3.58	0.32	88.9	-20π	8	1	0.63	10	10	2.0	1.0
D-2		0.55	8.01	3.22	0.26					0.50				
D-3		0.65	8.01	2.51	0.21					0.35				
N-1	Number of Whorls (n)	0.55	8.01	3.22	0.26	88.9	-10π	8	1	0.51	10	5	2.0	1.0
N-2							-20π					10		
N-3							-30π					15		
H-1	Cone Height (H_c)	0.56	9.01	2.80	0.36	88.9	-20π	9	1	0.65	10	10	2.0	1.0
H-2		0.5	8.02	3.58	0.32			8						
H-3		0.44	7.02	4.57	0.28			7						

Table B2

Values of parameters used in each model in the operating parameter studies. All linear parameters (H_c , R_c , R , A , d , d_p) are expressed in centimeters, angular parameters α , ω and β are expressed in degrees, θ_0 is expressed in radians and n is a number. Inlet velocity v is expressed in m/s. Note that the geometric parameters of the turritella were kept constant for studies involving the variation of the operating parameters. The geometric parameters were identical to that of Model D-1 in Table B.1.

Mod.No	Control Parameter	d_p	v	Slip Condition: Insert/Wall
P-1	Pipe Diameter (d_p)	2.0	1.0	NS/NS
P-2		2.05		
P-3		2.1		
P-4		2.2		
P-5		2.4		
P-6		2.6		
P-7		2.8		
S-1	Slip Boundary Condition	2.0	1.0	NS/NS
S-2				NS/CS
S-3				CS/NS
S-4				CS/CS
Va-1	Average Inlet Velocity (v)	2.2	0.5	NS/NS
Va-2			1.0	
Va-3			1.5	
Va-4			2.0	
Va-5			2.5	
Va-6			3.0	
Vb-1		2.0	0.4	
Vb-2			1.0	
Vb-3			1.6	
Vb-4			2.0	
Vb-5			2.5	
Vb-6			3.0	

Appendix C

Model equations of the shear stress transport (SST) Turbulence model

The finite element SST (Shear Stress Transport) model in COMSOL Multiphysics® is a Reynolds-Averaged Navier-Stokes model that combines the benefits of the superior behaviour of the k - ω model in the near-wall region, and the robustness of the k - ϵ model by interpolating between the two. The governing equations are given by (COMSOL, xxxx):

$$\rho \frac{\partial k}{\partial t} + \rho \mathbf{u} \cdot \nabla \mathbf{k} = P - \rho \beta_0^* k \omega + \nabla \cdot ((\mu + \sigma_k \mu_T) \nabla k) \quad (C1)$$

$$\rho \frac{\partial \omega}{\partial t} + \rho \mathbf{u} \cdot \nabla \omega = \frac{\rho \gamma}{\mu_T} P - \rho \beta \omega^2 + \nabla \cdot ((\mu + \sigma_\omega \mu_T) \nabla \omega) + 2(1 - f_{v1}) \frac{\rho \sigma_\omega}{\omega} \nabla \omega \cdot \nabla k \quad (C2)$$

where,

$$P = \min(P_k, 10\rho\beta_0^*k\omega) \quad (C3)$$

$$P_k = \mu_T \left(\nabla \mathbf{u} : (\nabla \mathbf{u} + (\nabla \mathbf{u})^T) - \frac{2}{3} (\nabla \cdot \mathbf{u})^2 \right) - \frac{2}{3} \rho k \nabla \cdot \mathbf{u} \quad (C4)$$

$$\text{Turbulent viscosity : } \mu_T = \frac{\rho a_1 k}{\max(a_1 \omega, S f_{v2})} \quad (C5)$$

The characteristic magnitude of velocity gradients :

$$S = \sqrt{2S_{ij}S_{ij}} \quad (C6)$$

Model constants are defined by : ϕ

$$= f_{v1}\phi_1 + (1 - f_{v2})\phi_2 \text{ for } \phi = \beta, \gamma, \sigma_k, \sigma_\omega \quad (C7)$$

where interpolation function f_{v1} and f_{v2} are given by:

$$f_{v1} = \tanh(\theta_1^4) \quad (C8)$$

$$\theta_1 = \min \left[\max \left(\frac{\sqrt{k}}{\beta_0^* \omega l_w}, \frac{500\mu}{\rho \omega l_w^2} \right), \frac{4\rho \sigma_{\omega 2} k}{CD_{k\omega} l_w^2} \right] \quad (C9)$$

$$CD_{k\omega} = \max \left(\frac{2\rho \sigma_{\omega 2}}{\omega} \nabla \omega \cdot \nabla k, 10^{-10} \right) \quad (C10)$$

$$f_{v1} = \tanh(\theta_2^2) \quad (C11)$$

$$\theta_2 = \max \left(\frac{2\sqrt{k}}{\beta_0^* \omega l_w}, \frac{500\mu}{\rho \omega l_w^2} \right) \quad (C12)$$

where l_w is wall distance.

The default constants used are given by:

$$\beta_1 = 0.075, \gamma_1 = 5/9, \sigma_{k1} = 0.85, \sigma_{\omega 1} = 0.5$$

$$\beta_2 = 0.0828, \gamma_2 = 0.44, \sigma_{k2} = 1.0, \sigma_{\omega 2} = 0.856$$

$$\beta_0^* = 0.09, a_1 = 0.31.$$

Model equations of the particle Tracing simulation

$$\mathbf{F}_D = \frac{1}{\tau_p} m_p (\mathbf{u} - \mathbf{v}) \quad (C13)$$

$$\tau_p = \frac{\rho_p d_p^2}{18\mu} \quad (C14)$$

where,

$\mathbf{F}_D$ is the drag force on the particle.

$\mathbf{u}$ is the velocity of the fluid.

$\mathbf{v}$ is the velocity of the particle.

m_p is the mass of the particle.

ρ_p is the density of the particle.

d_p is the diameter of the particle.

Appendix D

Details of Turbulent Flow Physics Interface in COMSOL®.

Parameter	Value / Setting
Working fluid	Water
Density of working fluid	From material (999.61 kg/m ³)
Dynamic viscosity of working fluid	From material (0.0010093 Pa.s)
Reference pressure	1 atm
Reference temperature	293.15 K
Turbulence model type	RANS
Turbulence model	SST
Wall treatment	Automatic

Appendix D (continued)

Parameter	Value / Setting
Discretization	P1 + P1
Turbulence model parameters	All set to default (as given in Appendix C)
Wall boundary condition	No slip (except for simulations S-2, S-3, S-4)
Solver configurations	Default
Relative tolerance	10 ⁻³

Details of Particle Tracing for Fluid Flow Physics Interface in COMSOL®.

Parameter	Value / Setting
Formulation	Newtonian
Wall condition	Freeze
Density of particles	1000 kg/m ³
Particle diameter	1 μm
Particle type	Solid particles
Release times	0 sec
Number of particles per release	3000
Drag law	Stokes
Turbulent dispersion model	None
Solver configurations	Default
Relative tolerance	10 ⁻⁵

Appendix E. Supplementary material

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.ces.2022.110831>.

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